

R. Brown



Dalhousie University

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Department of Psychology

23rd Annual
In-House Convention

28-29 April 1997

Monday, April 28, 1997 (LSC 4258/63)

8:00 - 8:15 **Coffee**

8:15 - 8:30 **Opening Remarks—Richard Brown**

8:30 - 10:00 **Session 1. Chair: Michael MacGregor**

- T1) S. Jerrott
- T2) R. D'Arcy
- T3) A.W. Lee and R. Brown
- T4) G. Gingras and D. Mitchell
- T5) S. H.-Greaven, D.A. Santor, R. Thompson and D.C. Zuroff
- T6) W. Brown

10:00 - 10:15 **Coffee Break**

10:15 - 11:30 **Session 2. Chair: Sherry Stewart**

- T7) D.P. Phillips, T.L. Taylor and S.E. Hall
- T8) J. Christie
- T9) S. Adamo
- T10) J. Fentress
- T11) S. Gadbois

11:30 - 12:45 **Lunch Break and Posters (Lounge):**

- P1) B. Winters and R. Brown
- P2) M.C. Dorris, T. Taylor, D.P. Munoz and R. Klein
- P3) R. Klein and F. Murphy
- P4) T. Taylor and R. Klein
- P5) B. Losier and R. Klein
- P6) F. Dastur, R. Brown, K. Harrigan, H. Wile and R. Brown
- P7) K. Skene and C. Moore

12:45 - 2:15 **Session 3. Chair: Vin LoLordo**

- T12) H. Schellinck and E.B. Keverne
- T13) R. Brown and L. Stanford
- T14) L. Stanford, J. Stapleton, P. Dill, A. Clarke, H. Randhawa, D. Dixon, A. Syed, Y. Mamoun and R. Brown
- T15) P.A. McMullen, A. Braithwaite and J.M. McGlone
- T16) M.G. Yoon and J.F. Connolly

2:15- 2:30 **Coffee Break**

2:30- 4:00 **Session 4. Chair: Brad Frankland**

- T17) S. Stewart, S.H. Taylor and J.M. Baker
- T18) A. Israeli and S. Stewart
- T19) M. MacGregor, K. Davidson and S. Gustys
- T20) J. Barresi
- T21) R. Klein and J. Hamm
- T22) S. Simpson, T. Trappenberg and R. Klein

4:00- 4:15 **Coffee Break**

4:15- 5:30 **Session 5. Chair: John Barresi**

- T23) C. Moore and M. Angelopoulos
- T24) S. MacLellan
- T25) E. MacLaughlin
- T26) S. Symons and H. MacLatchy-Gaudet
- T27) R. Brown and P. McGrath

There will be a cash bar social in the Lounge after the last session of the day.

Tuesday, April 29, 1997 (LSC 4258/63)**8:00 - 8:15 Coffee**

8:15 - 9:45 Session 1. Chair: Heather Schellinck
 T28) T.L. Fidler and V.M. LoLordo
 T29) C.A. Forestell and V.M. LoLordo
 T30) J. Emsley
 T31) M. deGroot and B. Rusak
 T32) C. Thompson, H. Westra, S. Stewart and A. MacDonald
 T33) D. Rioux

9:45 - 10:00 Coffee Break

10:00 - 11:30 Session 2. Chair: Margo Watt
 T34) R. Hoffman
 T35) W. Schmidt
 T36) L. Breau
 T37) D. Tripp, M. Sullivan and N. Neish
 T38) P.J. McGrath, C. Romus, C. Camfield, M.A. Campbell and A. Hennigar
 T39) W.K. Honig and B. Matheson

11:30 - 12:45 Lunch Break and Posters (Lounge):
 P8) R. Brown, T. Moran and B. Slotnick
 P9) R. Barnet, L. Corbit and V.M. LoLordo
 P10) J. McGlone, S.E. Black and A. Sita
 P11) M. MacGregor and D.A. Santor
 P12) R. Klein
 P13) W. Schmidt and R. Klein

12:45 - 2:00 Session 3. Chair: Paul Campagnaro
 T40) M. Watt, S.H. Stewart and B.J. Cox
 T41) A. MacDonald, M. Skinner, S. Stewart and B. Conrad
 T42) X.J. Sun and I.A. Meinertzhagen
 T43) B. Chen and S. Shaw
 T44) M. Shimohigashi and I.A. Meinertzhagen

2:00- 2:15 Coffee Break

2:15- 3:30 Session 4. Chair: Richard Brown
 T45) D.A. Santor
 T46) T. Trappenberg
 T47) S. Shaw and L. Varney
 T48) J.A. Horne
 T49) S. Stanley

3:30- 3:45 Coffee Break

3:45- 5:30 Symposium "Advances in Evolutionary Psychology"
 Chair: William Brown
 • Steve Shaw—Neural Evolution
 • Shelley Adamo—Darwinian Medicine
 • John Fentress—Gene-Environment Interactions: An Adaptationist Approach
 • Chris Moore—Evolution and Development
 • John Barresi—Evolution and Social Psychology

Open Discussion

There will be pizza and beer available in the Lounge after the last session of the day.

All talk sessions will be held in the LSC Rooms 4258/63.

T1. An Examination of the Long-Lasting Effects of Oxazepam on Implicit and Explicit Memory

Susan Jerrott

In the present study the effect of oxazepam on implicit and explicit memory was tested at two time points. Twenty university students were tested at 170 minutes after drug administration (close to the peak plasma concentration of oxazepam) and twenty students were tested at 240 minutes after drug administration (post-peak). At each time point, the subjects were randomly assigned to the drug condition (30 mg of oxazepam) or the placebo condition. Consistent with previous findings, oxazepam impaired performance relative to placebo on both the explicit memory test (cued recall) and the implicit memory test (word stem completion task) at 170 minutes post drug. Likewise, oxazepam also impaired explicit and implicit memory performance at 240 minutes post-drug. Oxazepam-induced memory impairments did not change relative to placebo from 170 minutes to 240 minutes post drug administration. Clinical implications of these long-lasting memory impairments will be considered. Although drug group effects were evident on attentional measures, (e.g. forward vs. backwards digits on the WAIS-R) these did not appear to be responsible for the observed memory effects.

T2. Assessment of language comprehension with event-related potentials

Ryan D'Arcy

The objective of the proposed research involves the assessment of receptive language comprehension with event-related potentials (ERPs). While there exists a panoply of electrophysiological investigations of language, recent studies (Byrne et al., 1995; Connolly et al., 1995; Ingles et al., 1996) have adapted standardized neuropsychological measures for computer presentation with ERPs. These investigations have demonstrated that ERP components (e.g., N400 and P300) can be elicited by standardized test stimuli and thus utilized to evaluate language functions in the absence of a behavioral response. Receptive language dysfunction is frequently noted in aphasic individuals suffering from a stroke, but is not always readily assessable with traditional psychometric measures due to limitations in motoric and communicative responses. Accordingly, the proposed investigation will adapt three prominent neuropsychological tests, the Token Test (De Renzi and Vignolo, 1962), sections of the Western Aphasia Battery (Kertesz, 1979), and a section from the Psycholinguistic Assessment of Language Processing in Aphasia (Kay, Lesser, and Coltheart, 1992). Performance on the computer tasks will be contrasted with the respective standardized tests for validation. It is hypothesized that ERP components will successfully differentiate behavioral responses for test items within an individual's ability, but not beyond. The results of this investigation, independent of the outcome, will provide insight into the neurological basis of language and the utility of ERPs in assessment.

T3. Effect of Male Removal on Gestation Length and Time of Parturition in Female California Mice (*Peromyscus californicus*)

Anna W. Lee & Richard E. Brown

California mice are a monogamous species with biparental care (Gubernick & Alberts, 1987, *J. Comp. Psychol.*, 101, 169-177). The presence of the male facilitates reproductive success by increasing pup survival and reducing the female's inter-birth-interval (Cantoni & Brown, *Anim. Behav.*, in press). The present study investigated the effect of male removal on day 32 of gestation on the timing of parturition. When compared to control pairs, females which had their mates removed had a longer gestation length and gave birth later in the day. Possible neuroendocrine mechanisms underlying the effects of the male on the reproductive physiology of the female California mouse will be discussed.

T4. Spatial localization in normal and amblyopic cats

Guy Gingras and Donald Mitchell

The main deficit in human amblyopia (blunt vision) concerns the coding of space and in particular, the relative localization of image features. This behavioral research project will study spatial localization tasks in normal and amblyopic cats using Gabor stimuli and a jumping stand in order to ascertain if the animal performance is similar to human data. Two parameters will be evaluated: the size and the contrast of the stimuli. The ultimate goal of the study is a better understanding of the neuronal mechanisms potentially responsible for the visual deficits described in human amblyopia.

T5. Is Self-Handicapping in Young Adolescent Girls Related to Mothers' Parenting?

Sonia H.- Greaven, Darcy A. Santor, Richard Thompson and David C. Zuroff

The use of self-handicapping strategies, which has been studied primarily in adults, may be related to the acceptance and reinforcement children experience while growing up. This study explored whether young adolescent girls' tendency to self-handicap was associated with their mothers' parenting characteristics. Mothers' ratings of care and stress were related to self-handicapping in their young adolescent daughters, as well as to their daughters' level of dysphoria. Regression analyses showed that higher maternal care was associated with lower levels of self-handicapping over and above the effect of their daughters' dysphoria, while mothers' stress from parent-child interactions was predictive of dysphoria in their daughters but not of self-handicapping.

T6. An Evolutionary Approach to the Signaling and Detection of Altruism

William M. Brown

In order for altruism between non-kin to be evolutionarily stable cheating must be detectable. However, the division of labour in many human cooperative ventures may render overt cheating undetectable. It is hypothesized that non-cheating individuals communicate their dispositions with hard-to-fake, nonverbal signals. Previous experiments have been consistent with this hypothesis (Frank, Gilovich, & Regan, 1993; Palameta & Brown, 1996). Video-clips of 4 altruists and 4 non-altruists were matched for physical attractiveness, expressiveness and content. One-hundred and twenty university students made open-ended personality assessments of the 8 targets. In addition, targets' facial micro-movements were coded by two blind raters. Results will be discussed in light of current conceptual models explaining the evolution of altruism between non-relatives.

T7. Son of Gap Detection: In One Ear and Then the Other

D.P. Phillips, T.L. Taylor and S.E. Hall

Last year, in fit of unbridled enthusiasm, we presented two lines of evidence that suggested that all was not as it appeared to be in the world of gap detection. Recall that gap detection is an auditory perception paradigm used to probe the temporal processes involved in speech recognition. It measures the shortest-duration silent period that is detectable between two noises. Historically, researchers have kept those noise markers identical -- which is not the case for the interesting gaps in speech. Last year, we showed that when the gap-detection paradigm is modified to mimic the speech signal (e.g., by making the noise markers spectrally dissimilar, so that they activate different perceptual channels), two different perceptual processes were engaged. One of these processes seemed to explain why there is a categorical boundary in the perception of voice onset times, and why it has the value that it does (about 35 ms). This year, we report an experiment in which we performed a new interrogation of this perceptual timing process.

T8. On the (ethereal, ephemeral) nature of negative priming

John J. Christie

Our data from spatial paradigms that typically generate Negative Priming (Tipper, Brehaut, and Driver, 1988) reveal that there may be important artifacts in unbalanced experiments of this kind: The costs in the ignored repetition condition are not due to special processing of the distractor. We subsequently conducted balanced and imbalanced designs in the identity task while removing potential artifacts from localization. We generated typical negative priming effects with an imbalanced design; but these effects disappeared when subjects no longer expected a target repetition event on the probe trial. Given that these basic paradigms used to study negative priming are brought into question it is likely that all of the paradigms that are believed to produce negative priming need to be re-examined. Additionally, the dramatic differences in the results of the spatial location and identity paradigms without biases may provide a platform for explaining differences between these two paradigms pointed out by others (Fox, 1995).

T9. Voices from within: Parasitic manipulation of host behaviour

Shelley A. Adamo

The parasitic wasp *Cotesia congregata* injects its eggs into the larvae of the tobacco hornworm, *Manduca sexta*. Hornworms were found to feed normally during the growth of the parasitic wasps. However, *M. sexta* stopped feeding several hours prior to the emergence of the wasp larvae. Hosts were not debilitated by the parasites and survived another 2 weeks without feeding after the wasps emerged. In the field, unparasitized *M. sexta* ate *C. congregata* pupae (n=10/10) suggesting that the suppression of host feeding prevents predation of the parasitic wasps by the host. The decline in host feeding correlated with a long-lasting increase in the concentration of octopamine in the host's hemolymph (22.2 ± 2.1 pg/ μ L (n=7) to 143.7 ± 7.8 pg/ μ L hemolymph (n=16)). Injections of octopamine depressed feeding in unparasitized *M. sexta*. The wasp larvae were not the source of the host's increased octopamine. Feeding was also temporarily depressed in *M. sexta* by injecting sephadex beads, bacteria (*Serratia marcescens*) or lipopolysaccharides (derived from *S. marcescens*). In all 3 cases, octopamine concentration in the hemolymph increased after injection (18.7 ± 4.2 to 54.7 ± 7.8 pg/ μ L hemolymph for bacteria-injected, n=10)). Feeding resumed with the decline in octopamine levels. Sephadex beads, bacteria, lipopolysaccharides and the emerging wasp larvae all activate the host's immune system. The data are consistent with the hypothesis that part of the parasite's ability to suppress host feeding involves permanently activating mechanisms that suppress feeding during an immune challenge.

T10. Perceptions of movement

John Fentress

Animal behavior is something of a mobile projective test, no matter how objective we try to be. What cues do we use, for example, to interpret the actions of other creatures? Part of the answer depends upon our measurement systems: "What you measure is what you get!" Of equal importance, measurements are inevitably incomplete and thereby can be filled-in through often undefined expectations, biases, and the like. We have thus launched a pilot project in which standardized behavior tapes and computer representations of movement are presented to people under different settings. The results can be scary (as in scare, not scar). It is clear that: a) cues can be used in an idiosyncratic manner, b) background events can change perceptions of behavior, and c) internal expectations of the observer can dictate what is recorded, and how these recordings are interpreted. Certainly the definition of objectivity must move beyond the position of shared bias. There may be a rich field for cognitive psychologists as well as ethologists to explore here. Suggestions are solicited.

T11. Behavioral sequences in canids: a new perspective on the plasticity of action and movement patterns

Simon Gadbois

Important dimensions of behavioral patterns may have been overlooked in the past because of (1) the species chosen, (2) the action patterns investigated and (3) the methods of analysis selected. The issues of intentionality (control) and automaticity, plasticity and stereotypy, stochasticity and determinism are reviewed in the light of new observations on caching sequences in canids. I will also discuss the pertinence of the concept of intensity patterns in behavioral sequences, which are often ignored to the profit of sequential patterns. An analogy with prosody in linguistics is suggested for the development of future models of temporal organization of behavior.

T12. Major urinary proteins of the strange male house mouse (*Mus musculus*) confer the specificity required to induce pregnancy block

Heather M. Schellinck¹ and E. Barry Keverne²,

¹Department of Psychology, Dalhousie University, ²Sub-Department of Animal Behaviour, University of Cambridge

The pregnancy of a female mouse (*Mus musculus*) is blocked by olfactory cues contained in the urine from a strange male but not in the urine from the stud male. We investigated the role of Major Urinary Proteins (MUPs) in determining these urinary cues by exposing pregnant female mice to one of the following: (A) urine from strange males; (B) chemically purified MUPs derived from strange male urine, with their ligands attached or (C) chemically purified MUPs derived from strange male urine, with their ligands removed via chemical extraction; (D) urine from stud males to which had been added chemically purified MUPs from strange males, with their ligands removed; (E) urine from stud males or (F) chemically purified MUPs from stud males with their ligands attached. (G) chemically prepared ligand, 2-(sec-butyl)thiazoline. Statistical analysis revealed that the exposure of pregnant females to (A) urine from strange males, (B) strange MUPs either with or (C) without the ligand attached or (D) combined with stud urine blocked significantly more pregnancies than exposure of female mice to (E) stud urine or (F) MUPs and ligand extracted from the urine of stud males or (G) chemically prepared ligand. These results indicate not only that (1) the MUPs of male mice demonstrate pheromonal activity whereas the volatile ligands do not but also that (2) the MUPS of strange males confer the specificity required to induce pregnancy block in females.

T13. The Hebb-Williams Maze: Fifty years of research (1946-1996)

Richard Brown & Lianne Stanford

The Hebb-Williams (H-W) maze was designed as a test of animal intelligence and since 1946 has been used to study the learning abilities of a wide variety of species including fish, raccoons and cows, as well as mice and rats. The H-W maze is an open field, with a start box, a goal box and moveable barriers. Each animal runs through a battery of twelve mazes of increasing difficulty, created with the movable barriers. As well as comparative studies of learning using different species, the H-W maze has been used to assess the effects of environmental enrichment or deprivation on learning and the impact of hormones, X-irradiation, specific brain lesions, changes in diet, and a variety of drugs on learning. Currently, the H-W maze is used to study genetic influences on learning and the effects of prenatal manipulations of diet, stress, hormones and drugs on learning, as well as the neural basis of learning. The present talk discusses the value of the H-W maze for studying advanced cognitive tasks and its potential as a tool for measuring "intelligence" in congenic strains of mice, transgenic mice and knockout mice.

T14. Measuring learning in CD1 mice in the Hebb-Williams Maze: Practical problems in the use of a complex maze test of intelligence

Lianne Stanford, Jennifer Stapleton, Patti Dill, Alison Clarke, Harleen Randhawa, Deana Dixon, Asma Syed, Yasmin Mamoun & Richard Brown.

Although the Hebb-Williams (H-W) maze was developed as "a method of evaluating animal intelligence", the procedures used for testing subjects in this maze have varied greatly. The most commonly used procedure is that developed by Rabinovitch and Rosvold (1951, *Can. J. Psych.*, 5, 122-128). As we plan to use this maze to study learning in different mouse strains, it is important to develop a standardized procedure. Using data from pilot studies with CD1 mice, this talk will focus on maze design, floor patterns, scoring procedures, motivation and all of the little things that can go wrong. We will also include some video footage of mice learning the H-W maze.

T15. Separate Orthographic Input and Output Lexicons: Evidence from a Surface Dysgraphic

P.A. McMullen, A. Braithwaite, and J.M. McGlone,

A surface dysgraphic was investigated to determine: 1). his writing deficit within a cognitive model and 2). support for a single or separate input and output orthographic lexicons. Regular words and nonwords were spelt perfectly, unlike irregularly spelt words (66% errors). All of these words were read perfectly. These results, along with normal cross-case transcription (e.g. CaSe- cAsE) supported intact orthographic input and output lexicons. Written rhyme fluency and deciding if words that sounded the same were spelt the same (e.g. tough-buff) was poor. Results were most consistent with a disconnection between phonological and orthographic output lexicons.

T16. Influences of sentential contexts on event related brain potential responses to semantic anomaly and/or verb inflectional disagreement in compound sentences.

Myong G. Yoon and John F. Connolly

We studied the characteristic patterns of neural responses of the human brain to various combinations of semantic anomalies and/or verb inflectional disagreements during rapid reading of compound sentences by analyzing the event related brain potentials (ERP) recorded from normal adult subjects. [1] ERP of semantically anomalous verbs showed negative shifts peaking at about 400 msec (N400) with respect to the semantically appropriate verbs. The onset time of the N400 responses depended on the syntactic contexts of the verbs. [2] ERP of inflectionally incorrect verbs made positive shifts at about 600 msec (P600) with respect to control verbs. A repetition of the inflectional disagreement in the second clause diminished the P600 response. The result suggests that the repeated inflectional disagreement tends to be ignored by the neural mechanism which has responded to the previous inflectional disagreement as P600. [3] The predominant component of neural responses to semantically anomalous and inflectionally incorrect verbs was P600 rather than N400. Thus inflectional disagreement (rather than semantic anomaly) seems to be a predominant linguistic feature, detected by those neurons whose activities happened to be monitored by our recording methods. [4] Inter-sentential influence of the semantically anomalous transitive verb (with respect to its subject) upon its (acceptable) object; ERP to the object words of semantically anomalous verbs showed large negative shifts during a period between 300 and 750 msec. The magnitude and duration of the negative shift in the object word was far greater than that in the anomalous verb itself. The result reveals the prominent influence of the two different sentential contexts on the same terminal words. [5] Inter-sentential influence of inflectional disagreement of semantically normal transitive verb upon its appropriate object in the second clause; ERP to the object words of the inflectionally incorrect normal verbs showed large negative shifts during a period beginning 300 msec. This large negative shift in the object words reflects the influence of the preceding inflectional relations between the subjects and the verbs in the compound sentences.

T17. Gender differences in anxiety sensitivity dimensions

Stewart, S. H., Taylor, S., & Baker, J. M.

The present study examined whether there are gender differences in the hierarchical factor structure of the Anxiety Sensitivity Index (ASI), and/or patterns of ASI factor scores. Principal components analyses on the ASI items of male (N = 290) and female (N = 528) students revealed nearly identical lower-order factor structures across genders (i.e., physical, psychological, and social concern factors). Nearly identical unidimensional higher-order solutions were also found across genders. Analyses on ASI lower-order factor scores showed that: females scored higher than males only on the physical concerns factor; females scored higher on the physical concerns factor than on the psychological and social concerns factors; and males scored higher on the psychological and social concerns factors than on the physical concerns factor. Females also scored higher than males on the higher-order ASI factor. The study provides support for the conceptual distinctions between the three lower-order ASI factors, as well as the first empirical demonstration of a hierarchical structure for the ASI. Men and women also appear to differ on AS dimensions in ways consistent with sex role socialization practices.

T18. Memory bias for food information in restrained eaters

Anne Israeli and Sherry Stewart

Research on cognitive patterns in individuals high in dietary restraint indicate an attention bias toward food related information (see Francis, Stewart, and Hounsell, in press). The current study was conducted to see if restrained eaters show a memory bias for food words in comparison to animal words (control list) in relation to individuals low in restraint. Thirty females (Mean age = 21.50, SD = 2.97) encoded a list of words (15 food, 15 animal) presented in an incidental learning task. Participants were then asked to freely recall as many words as they could remember and fill out questionnaires including the Restraint Scale (Herman & Polivy, 1980). Participants were classified as high in restraint (n=16) and low in restraint (n=14) on the basis of the scale. Preliminary analyses indicate that individuals high in restraint remember more food than animal words and individuals low in restraint do not demonstrate such a bias. The results support (Bemis-Vitousek & Hollon's (1990) theory that restrained eaters have highly organized cognitive structures (schema) which permit them to avoid dangers (food) in their environment.

T19. A translation of borderline and histrionic personality disorders

Michael MacGregor, Karina Davidson and Stephanie Gustys

This paper addresses the translation of Borderline and Histrionic personality disorders into the language of the Five-Factor model of personality. Based on theoretical expectations, we investigated the empirical relationship between the Five-Factors of personality and these two personality disorders in a non-clinical sample. As expected, the personality disorders were characterized by high levels of Neuroticism. For Borderline personality disorder, our expectations regarding Agreeableness were confirmed, however, those for Extraversion were not. As well, additional relationships were found between both Openness and Conscientiousness and Borderline personality disorder. For Histrionic personality disorder, our expectations for Extraversion were confirmed, however, those for Openness were not. Additionally, both Agreeableness and Conscientiousness demonstrated an unexpected relationship with Histrionic personality disorder. While some of the expected relationships between the Five-Factors and Borderline and Histrionic personality were confirmed, others were not. It appears that theoretical or clinical expectations are not necessarily confirmed by empirical findings in a non-clinical sample. Results are discussed in terms of the utility of the Five-Factor model in describing personality disorders.

T20. Becoming conscious of 'consciousness': Notes for a history of an important psychological concept

John Barresi

During the 17th century the term of 'consciousness' suddenly appeared as a psychological concept. Since then this concept has become increasingly important both in the foundations and later developments of 'mental philosophy' and scientific psychology. Although during the heyday of behaviourism it diminished in importance, interest in consciousness has been revived again toward the end of this century. In the present talk I will look at the origins of the use of this term, its ancient sources in Greek and Latin, its major 'discovery' in Descartes, and its later development in the works of Cudworth, Locke, Leibniz, Hume, and others in the late 17th and early 18th century.

T21. Attention (to the moving line): "Where you lead, I will follow".

Raymond M. Klein & Jeff P. Hamm

When a line is presented in the vicinity of a recent luminance change (peripheral cue) it is perceived to be drawn over time away from its "cued" end even though the line is actually presented all at once. This study was designed to determine if attention, exogenously attracted to the cue, would come under the exogenous control of this illusory motion and follow the drawn line from its cued end to its terminus. Each trial began with the display of 4 small squares at the corners of an imaginary square centered about fixation. After 750 ms one of the 4 squares brightened briefly (50ms) in the cued conditions. Simultaneous with the offset of the cue a horizontal line was presented joining the upper or lower boxes. After 100 ms one of the 4 squares was transformed into a horizontal or vertical rectangle for 150 ms. The subject's task was to make a horizontal/vertical decision as rapidly as possible. The different events (cues, lines and targets) were combined factorially so that neither cues nor lines were informative about target location. Responses to targets at an uncued location that was at the end of an illusorily drawn line were as fast as to targets in the neutral and "valid" conditions, and much faster than the remaining invalid conditions, strongly suggesting that attention follows the motion in the "shooting line" illusion.

T22. Integration of Environmental and Intentional Information within the Saccadic Programming System: A Computational Perspective.

Sol Simpson, Thomas Trappenberg, & Raymond Klein

Saccadic reaction time (SRT) is influenced by both the observer's environment (exogenous factors) and intentions (endogenous factors). To investigate these two SRT contributions, the time interval between the removal of a fixation point and the presentation of a peripheral target (the Gap Paradigm) as well as the probability of a target appearing at a given location were varied. It was hypothesized that if endogenous information affects SRT within the same functional layer of the saccadic programming system that is responsible for exogenous control (the superior colliculus), the effect of target location information would be significantly larger in conditions where the fixation stimulus was removed prior to target presentation than in conditions where the fixation stimulus remained present following target onset. This interaction was obtained with saccadic but not manual RTs, suggesting that the lateral inhibition of fixation cells in the rostral pole (RP) of the superior colliculus (SC) onto the saccade related cells of the SC may mediate the effects of both exogenous and endogenous inputs on SRT. This presentation will attempt to explain this interaction using a computational perspective within which a neural field model of saccadic programming is constrained by recent neurophysiological data.

T23. Word learning in the context of referential and salience cues

Chris Moore & Maria Angelopoulos

Previous research has shown that word learning can occur in the context of joint attention, however it remains unclear whether such learning occurs because the child understands the referential intent of the speaker or because the speaker's referential behavior tends to make the intended object more salient. In this work, both referential and salience cues were independently manipulated in a study of novel word learning. Eighteen- and 24-month-old children were presented with a novel word in the context of two toys, which could be activated by remote control. In the match condition, the toy which was the target of the adult's referential behavior was activated while the other was not. In the mismatch condition, the toy which was the target of the referential behavior was not activated while the other was. Two control conditions had either both toys simultaneously activated or neither toy activated while the adult made reference to one.

Comprehension and production posttests assessed the extent of acquisition of the novel word. The results showed that at 24 months children tended significantly to pick out the target toy in response to the novel word under all conditions and that there was no significant difference across conditions. There was also no difference across conditions in children's ability to produce the novel word appropriately. At 18 months, however, there was a difference in performance across conditions. In the mismatch condition, the children were more likely to pick out the activated toy in response to the novel word, while in the other three conditions, they were more likely to pick out the target toy. These results imply that word learning in joint attention situations may initially occur because of the salience of the target object but soon does occur because the child understands the referential intent of the speaker.

T24. What does that Robot want from me now? An investigation of young children's abilities to repair contingent queries.

Suzanne MacLellan

The ability of 2-year-old children to respond differentially to specific and neutral contingent queries was examined with a toy robot as the communicative partner. 22 children (mean age = 2;9 years) were asked contingent queries immediately following 10 of their utterances in a free play session. 5 specific queries (ie. "That's a what?") and 5 neutral queries (ie. "What?") were asked to each subject. Children's replies were coded to determine the type of response strategy used: repetition; specification; or reformulation. Results revealed that children were significantly more likely to specify part of their original utterance following a specific query, whereas following a neutral query they were significantly more likely to repeat their entire utterance. No differences were observed in the number of reformulations made. Results will be discussed in terms of the young child's developing theory of mind. Additional research being designed as a follow up to these results will also be discussed.

T25. Inhibition as the Primary Deficit in ADHD

Elizabeth MacLaughlin

Researchers are continually trying to isolate a core deficit in children with attention deficit hyperactivity disorder (ADHD). Recently, Barkley (1997) proposed that behavioral inhibition was the primary deficit, and that it set the stage for the observed deficits in executive function and response execution in children with ADHD. Pure behavioral measures of inhibition, however, may be confounded by socialization, and offer little in the way of an explanation of the mechanism(s) underlying the behavioral disturbances. Cognitive or attentional inhibition may be a more reliable and valid construct for study. Deficits in this area correspond with evidence suggesting that ADHD may result, in part, from frontal lobe dysfunction. Research on cognitive inhibition in children with ADHD will be reviewed, and further studies will be proposed that could validate and enhance Barkley's model.

T26. Teaching Children How to Locate Information: Strategy Instruction for Students in the Elementary School Grades

Sonya Symons, Heather MacLachy-Gaudet

The present study examined the effects of instructing elementary school students how to most efficiently locate information in a children's encyclopedia. The instructional protocol emphasized the importance of using the index, reading the text carefully and self-monitoring. 155 students from grades 3, 4, and 5 were randomly assigned to one of three groups: Strategy for Locating Information (SLI), Strategy for Locating Information without Monitoring (SLI-M), or Non-Instructed Control (CON). Although students in both the SLI and SLI-M groups found more correct answers than those in the CON group, the monitoring component was important in improving search efficiency (as measured by number of correct answers found per minute). Students who received instruction were more likely to use the Index than other students, and they also searched fewer total pages than the CON group.

T27. The National Conference on Psychology as a Science

Richard Brown and Pat McGrath

From 8-11 May, 1997, CPA and BBSCS have invited 45 Canadian psychologists to a conference to discuss the future directions in MRC, NSERC and SSHRC funded research. The goal of the National Conference is to "develop a vision of research and training in psychology that is in tune with the increased realization that many of our Society's most pressing needs and exciting opportunities will benefit directly from accurate and sophisticated knowledge of human thought, emotion and behavioural processes". "We must identify research priorities that will allow psychology to make a meaningful contribution to the national research agenda, as described in a recent report on Science and Technology entitled 'Healthy, wealthy and wise: A framework for an integrated federal science and technology strategy'." In brief, the question that we have been posed is: "Whither psychology in the 21st century?"

T28. Does manipulation of thirst change ethanol consumption by rats anticipating footshock?

Tara L. Fidler and Vincent M. LoLordo

Rats were given the opportunity to drink ethanol, sucrose or water in a distinctive environment prior to shock sessions in a third environment. Since thirsty rats will learn to consume otherwise aversive fluids when there is no other fluid available we wanted to show that our rats were motivated to consume ethanol for reasons in addition to thirst. Therefore on some days the rats had an opportunity to consume water in their home cages prior to being placed in the drinking environment. Preliminary results appear to show that water consumption in the drinking environment is more affected by this manipulation than is consumption of ethanol or sucrose. However, fear in the shock box (as measured by bolus production) appears to be differentially affected in the ethanol and sucrose groups.

T29. Pitfalls in odour and taste preference conditioning

C.A. Forestell and V.M. LoLordo

The standard procedure used to condition flavour preferences involves a within subject differential conditioning paradigm in which the cue flavours and the appropriate reinforcers are presented simultaneously. In the ensuing test phase, animals typically show a robust preference for the CS+ which does not readily extinguish. The present research, which investigated the ability of a taste to potentiate the formation of a conditioned odour preference has necessitated several modifications to this standard procedure. These included the use of a reverse order conditioning procedure adapted from Boakes & Lubart instead of a simultaneous procedure, between subjects rather than within subject comparisons, and animals which received restricted rather than ad lib water access. Each of these procedural modifications will be assessed in terms of its influence on our failure to obtain robust odour and taste preferences.

T30. Anatomical and Neurochemical Characterization of the Development of the Embryonic Brain of *Drosophila melanogaster*

Jason G. Emsley

Current and proposed research consists of three inter-related projects. Current work involves the production of a detailed, three-dimensional anatomical map of the developing embryonic brain and ventral nerve cord (vnc) of *Drosophila*. This map will be a detailed template onto which information about cell lineage, gene expression, neurotransmitter expression, and biochemical activity can be mapped. The second phase focusses on localization of serotonergic expression in the embryonic brain and vnc. The expression of nitric oxide (NO) during embryonic neural development will also be characterized. NO is a putative neurotransmitter which has been implicated in neuronal development, death, and plasticity.

T31. A Little Light Learning?

Marleen de Groot and Ben Rusak

The circadian system responds directly to biologically relevant environmental stimuli such as the light-dark cycle. A recent study (Stewart & Amir, 1996) showed that an external stimulus that does not induce phase shifts or expression of the immediate-early gene product *c-fos* in the SCN of rats can be made to do so after being paired repeatedly with a phase-shifting light pulse in a Pavlovian conditioning paradigm. Using appropriate conditioning and control procedures, we are attempting to assess whether this phenomenon is reliable and occurs in both rats and hamsters. Animals were housed in isolation boxes containing an activity wheel, a light source and a fan. Wheel-running activity was monitored throughout the study, and was used as the behavioural measure of circadian phase. The animals were entrained to a standard light dark cycle, and conditioning took place during the dark phase of the cycle. The unconditioned stimulus (US) used was a fifteen minute light pulse, and a twenty minute air disturbance served as the conditioned stimulus (CS). Results of our initial studies will be described.

T32. Relations between Benzodiazepine Use, Agoraphobic Avoidance, and Autobiographical Memory in Clinical Anxiety

Carol Thompson, Henny Westra, Sherry Stewart, and Alan MacDonald

The purpose of this study was to investigate overgeneral reporting of past personal events in clinically anxious patients, as a function of benzodiazepine (BZ) use and avoidance of fearful situations. Using the Autobiographical Memory Test, BZ-using anxiety patients, nonmedicated anxiety patients, and controls were asked to recall specific personal memories in response to cue words which varied in emotional valence. Results showed that, among BZ-using patients, more overgeneral memories were recalled in response to negative cues as cumulative BZ dose increased. A positive relationship between overgeneral recall and a tendency to avoid anxiety-arousing situations (as measured by the Mobility Inventory) was also found in the clinically anxious groups but not in the control group. These results suggest that, among clinically anxious patients, overgeneral reporting of past personal events is related to BZ dose and to a tendency to avoid anxiety-arousing situations. Clinical implications of these findings will be discussed.

T33. Are you feeling childish about your present concerns? Some reasons why.

David Rioux

A narrative research methodology developed by H.J.M. Hermans (1992) is being used to investigate relationships between peoples' earliest memories and their present concerns which have been postulated by Adlerian and Cognitive-Perceptual (CP) theories. Adler viewed earliest memories as relatively constant reflections of one's lifestyle, while CP theory predicts shifts in earliest memories with changing present concerns (Bruhn, 1992). Opposing predictions based on these two hypotheses were tested in an experiment involving forty-five students who participated in two sessions separated by a month. In each session they provided narrative descriptions of six earliest memories and six present concerns as well as self-rated affective profiles for each of these life experiences. Comparisons of life experiences and affect profiles over time will address the issue of permanance among earliest memories, where Adler's theory would predict one's very earliest memory to be the same across two sessions and relate more strongly to present concerns than would other memories, and CP theory holds that all early memories are subject to change along with changing present concerns.

T34. Footprints on the Internet: What do they tell us?

Ron Hoffman

The Psychology Department home page officially went on the World-Wide Web on November 29th, 1996. The site lives on a server provided by the University Computing and Information Services, and a log is kept of all visits to the pages in the site. Every time a page is visited, information is logged about the visitor. In this talk, we'll have a look at what these "footprints" in the log tells us about who has visited the site, where they come from, and what gets looked at. Secret features of the Psychology site will be revealed. The feedback features of the site will get special attention.

T35. Surveys and Tests on the Web: A Tutorial

Billy Schmidt

As many members of the department are aware, I have created a tool that allows non-programmers to implement surveys and tests electronically on the world-wide web. The tool is freely available to members of the department, and I would like to see it used to benefit researchers at Dalhousie, regardless of whether they use it to collect data from the web populous or internally from subject pool participants. I will demonstrate the creation of a web survey from start to finish, and suggest possible uses within the department.

T36. Predicting Children's Pain

Lynn Breau

This study examined the pain experienced by 112 children aged 4 to 6 receiving routine DPT immunization. Pain during the immunization was measured using the Child Facial Coding System (Chambers, et al., 1996). The Child Facial Coding System is an observational scale used to record the intensity of 13 facial actions believed to be indicative of pain in children. Ratings of one second intervals were made from videotape over a 30 second period which included: 10 seconds prior to injection, 10 seconds during injection, and 10 seconds after injection. After immunization, each child, their parent, and an observer rated the child's pain on a Faces Pain Scale. The adults also rated the child's pain on a Visual Analogue Scale. The results indicate that post-immunization pain ratings were related to: the informant, the sex of the parent, the age of the child, whether the child was on medication during the previous month, whether they were currently on medication, and whether they had previous experience with needles. Principal components analysis of the 13 facial actions resulted in four factors common to all three periods: horizontal mouth movements, open mouth, blinking, and eye squeezing. The relative importance of these factors, however, varied by time period. Finally, different combinations of facial actions and medical history were found to predict post-immunization pain ratings, depending on the informant and the measure used.

T37. What Hurts and What Helps During Dental Hygiene Treatment?

Dean Tripp, Michael Sullivan and Nancy Neish

Given the lack of research quantifying pain during various dental hygiene procedures, pain and pain coping attempts were explored in 48 patients (27 male, 21 female) undergoing routine dental hygiene treatment. Dental anxiety and self-efficacy of pain coping attempts were also explored in relation to pain. The majority of patients (79%) report none or very little pain during clinical procedures. In direct contrast, 21% of sample reported pain scores that were extreme in comparison to the majority. Empathic discussions with the hygienist, self-distraction from the pain, and information seeking from the hygienist, were the most often employed pain coping strategies. In dividing the sample by pain reports (i.e., low and high pain), no group differences in regards to dental anxiety, age, education, brushing habits, or the degree of treatment difficulty were found. However, the low pain group flossed more ($t=3.01$, $p<.005$) and had greater levels of self-efficacy for pain coping attempts ($t=2.28$, $p<.027$) than did the high pain group. There were no differences in how many pain coping strategies were reported between low and high pain groups as well as low and high self-efficacy for pain coping. Results suggest that feelings of personal control during dental hygiene treatment may be a factor in successful pain management. Future research is discussed in terms of other trait-like cognitive factors such as pain catastrophizing.

T45 T38. **Behaviors caregivers use to determine pain in non-verbal, cognitively impaired children**

Patrick J. McGrath, Christina Romus, Carol Camfield, Mary Ann Campbell, Allan Hennigar

Primary caregivers of 20 children (age=14.5 years and language level=10 months) were recruited and interviewed to recall 2 instances of short sharp pain and 2 of longer-lasting pain and describe their child's behaviour. Transcribed interviews were reviewed and sets of non-overlapping items were developed. Thirty one behaviours were extracted from the interviews. The specific behaviors were often different from one child to another but the classes of behaviors (Vocal, Eating/Sleeping, Social/Personality, Facial Expression, Activity, Body & Limbs and Physiological) were common. Reliability of using the checklist on interviews was good ($\kappa=.77$). The checklist has excellent content validity and will be useful for caregivers of cognitively-impaired, non-communicating children to report on pain behaviors.

T39. **Transposition of an Intermediate Numerosity Discrimination**

W. K. Honig & W. R. Matheson

Pigeons were trained to respond differentially to an array with a relative numerosity (proportion of red to blue elements) intermediate to that of two other arrays in a successive conditional discrimination, and were then transferred to different sets of numerosities. The mechanism of transfer appeared to be absolute, rather than relational. The procedure was then changed to a simultaneous forced choice of the intermediate numerosity among three arrays. Performance within sessions suggested the operation of a relational process. The apparent discrepancy between the two results, and the relative influence of numerosity and entropy, will be discussed.

T40. **Learning Experiences in the Development of Anxiety Sensitivity**

Margo C. Watt, Sherry H. Stewart, and Brian J. Cox

The present study involved a conceptual replication of Anke Ehlers' (1993) work on panic and childhood learning experiences, but also extended her work by investigating the relationship between these learning experiences and the development of anxiety sensitivity (AS). Five hundred and fifty-one students participated in a retrospective assessment of their childhood and adolescent learning experiences with respect to somatic symptoms. Panic-reporters in the study revealed a pattern of learning history results similar to Ehlers', however, a different pattern emerged in the learning histories of highly anxiety sensitive individuals (HAS). Unlike panickers, and contrary to predictions, the learning experiences of HAS individuals were found not to be specific to anxiety symptoms but involving parental reinforcement and modeling of sick-role behaviour related to somatic symptoms in general. HAS subjects reported both more anxiety and cold symptoms prior to age 18 than individuals with lower levels of AS. In addition, both cold and anxiety symptoms elicited special attention and/or instructions for HAS individuals to take special care of themselves. These findings are contrasted with the results for panickers in the sample who reported learning experiences (modeling and parental reinforcement) specific to anxiety-related symptoms. Purported links between AS, panic, and hypochondriasis are discussed.

Whitehead - menstrual disorders

T41. A Dose-Response Study of the Effects of Alcohol on the Response to Hyperventilation in High and Low Anxiety Sensitive Participants

Alan MacDonald, Michelle Skinner, Sherry Stewart, Brent Conrad

Previous research suggests that high levels of anxiety sensitivity (AS; fear of anxiety symptoms) may constitute a risk factor for alcohol abuse. The present study evaluated the hypothesis that high AS levels may heighten risk for alcohol abuse by increasing a person's sensitivity to the stress response dampening (SRD) effects of alcohol, which could negatively reinforce drinking in this population. Additionally, the study investigated whether relatively high doses of alcohol are needed in order to see significant SRD effects among high AS individuals and thus explain their more frequent drinking to the point of legal intoxication. Seventy-seven student participants (39 high AS; 38 low AS) were randomly assigned to one of three beverage conditions: placebo, low dose alcohol, and high dose alcohol. Following beverage consumption and absorption, participants underwent a three minute voluntary hyperventilation challenge to induce bodily arousal symptoms. High AS placebo participants reported greater affective and cognitive reactivity in response to the challenge than low AS placebo participants, indicating increased fear and catastrophising during hyperventilation among sober high AS individuals. Both the high and low doses of alcohol eliminated the differences in responding between AS groups by normalizing the affective and cognitive reactivity of the high AS students to the challenge. However, only high AS subjects administered the high alcohol dose (blood alcohol concentration (BAC) = 0.08) showed reduced cognitive and affective reactivity to the hyperventilation challenge when compared to high AS subjects administered placebo. Finally, all subjects (high and low AS alike) in both alcohol conditions showed reduced somatic reactivity to the challenge compared to subjects in the placebo condition. Clinical and theoretical implications of the findings are discussed.

T42. Looking through the fly's brain - application of confocal laser scanning microscopy to the study of *Drosophila* development.

X.J. Sun and I.A. Meinertzhagen.

Combined with powerful digital imaging techniques, confocal laser scanning microscopy (CLSM) provides unprecedented views of biological specimens. We present here some recent CLSM data to study the development of the brain in the fruitfly *Drosophila melanogaster*. Using antibodies against the synaptic vesicle protein synaptotagmin, and various other markers, synaptic regions and specific cell types in *Drosophila*'s brain are readily revealed in three dimensions throughout development. This technique reveals new images of, and insights to, the functional architecture and make-up of the brain in this genetically manipulable animal. The use of this technique in genetic screens will be shown by CLSM analyses of structural brain mutants.

T43. Electrical- and dye coupling between presynaptic photoreceptor axons in fly visual system

Bin Chen and Steve Shaw

Photoreceptors from different facets of the fly's compound eye converge in stereotyped groups of 6, the cylindrical synaptic modules ("cartridges") that make up the first synaptic zone behind the eye (lamina). There they make about 1200 histaminergic synapses to postsynaptic neurons, in big flies. In addition, electronmicroscopy shows that each pair of terminals in the cartridge is interconnected presynaptically by about 30 small maculate gap junctions each, raising the expectation that the 6 contributing photoreceptor axons to a cartridge should be strongly electrically coupled (Shaw & Stowe, 1982). We will briefly review older evidence (Shaw 1984, van Hateren 1986) that the terminals are indeed electrically coupled, when probed optically to stimulate cells from the six different facets of the 6-way convergence. The results allow the degree of coupling to be estimated, confirming that it is indeed strong.

When neurons have been found to be electrically coupled with gap junctions in other neural systems, injections with small fluorescent tracers have often showed them to be "dye-coupled" as well. Lucifer Yellow (MW 457) is small enough to pass through the central connexon channel of some gap junctions. Unfortunately, many tests so far (Hardie, Shaw) have completely failed to detect dye-coupling between photoreceptor terminals. This has now been repeated with the slightly smaller, more recently introduced tracer, biocytin (MW 372). After many test injections, biocytin has been found to spread to either one or two other terminals in the same cartridge group, always on a next-neighbour basis. This finally confirms the other evidence, and suggests that coupling strength is greatest between next-neighbour cartridge terminals, as expected.

T44. The gene *shaking-B* (*Passover*) regulates the number of gap junctions in the visual system of *Drosophila*: the jump between behaviour and molecules

M. Shimohigashi¹ and I.A. Meinertzhagen

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Gap junctions are the structural contacts that mediate electrical coupling between neurons in nervous systems. In vertebrates, gap junctions form from connexin proteins, but in invertebrates the corresponding proteins remain elusive. Recently, however, the *shaking-B* gene in *Drosophila* is suggested to encode a connexin-like protein, and mutants of *shaking-B*, including *Passover*, a null mutant with a defective escape jump, exhibit coupling defects. An antibody raised against the *shak-B* peptide expresses in the the lamina, where terminals of the eye's photoreceptors are electrically coupled by gap junctions. Compared with the wild type, there are significantly fewer (ca 20%) gap junctions in the *shak-B* mutant, but only in the upper levels of the lamina. At deeper levels, the numbers are smaller in both mutant and wild type, and both the same. Thus, the lamina's population of gap junctions is partitioned into at least two strata, and the numbers in the mutant imply the existence of at least two genetic factors, possibly proteins, encoding the number of gap junctions.

138 T45: **Neural Statistics, Human Failing, and the Backward Propagation of Error**

D.A. Santor

This brief talk introduces the basics of neural networks and machine learning and illustrates how for some (if not many tasks) we're better off not relying on human judgement.

T46. **Recurrent network dynamics**

Thomas Trappenberg

The brain does not process information in a strictly linear fashion. It has rather a large amount of recurrent pathways which we believe are the root of its rich dynamical behavior. In this talk we will discuss some properties of very simple recurrent abstract neural networks. We will demonstrate their behavior with some simulations.

T47. **The evolution of the blood-brain barrier in the insect eye: is there a barrier in Archaeognatha?**

Steve Shaw and Lesley Varney

In vertebrates, the blood-brain barrier (BBB) is located in the brain capillary walls and stabilizes the brain's micro-environment, protecting it from potentially harmful incursions of blood chemicals, etc. The evolution of the barrier mechanism is not easy to study by comparative methods, since all known vertebrates have it. The other larger group of terrestrial colonizers, winged insects (Pterygota), also have a BBB with the same role at the surface of the CNS, and which extends out into the eye. In trying to trace its evolution, we have looked to see if the eye barrier is present in the more primitive wingless insects (Apterygota), from which pterygotes originally evolved back in the Paleozoic era.

We collected Petrobius maritimus (jumping bristletails) of the most primitive extant apterygote family Machilidae (Archaeognatha) on the shore near Parrsboro. Compared to other apterygotes, they have very well developed compound eyes. A colony of about 40 has been kept successfully for about 6 months in the lab. Using microelectrodes to pressure-inject small fluorescent tracers into the haemolymph (blood) of the head, we observed that these dyes lit up the eyes within a few minutes of injection. Specimens fixed and sectioned showed that dye is indeed present in the eye, concentrating particularly in the outermost optical layer, the cornea, which could only have reached by diffusing up from the base of the eye. We are currently setting up to check this, using electron-microscope tracers. Next, bugs were injected on a tilt stand (goniometer) mounted directly on the microscope, allowing fluorescence emission from part of the eye surface to be measured optically in real time with a sensitive detector. The measured time profiles confirm that dye uptake is extremely rapid ($t_{1/2} \approx 0.4 - 1.4$ min), although it then disappears again from the eye. Finally, the path-length for diffusion into the eye has been measured and the expected diffusion profile modelled, from a solution to the diffusion equation for the case with no barrier. This predicts similar values of $t_{1/2}$ (0.8 - 1.1 min) without further assumptions, confirming that the eye space must be completely open to the circulation. The situation in flies (advanced pterygotes) is quite different, with very slow penetration and loss of dyes. The key reason for the lack of a barrier in machilids may be the primitive stage of evolution of the tracheal system in the group, which needs upgrading before barriers cutting off the retina become feasible.

T48. Digitize that Image

Jane-Anne Horne

This talk will introduce the process of digitizing images and the subsequent techniques that can be used to analyze or process a digitized image. I will give a brief review of some of the techniques that have been successfully used in recent years, with some insights on what a digitized image is and some hints on getting the best image possible. I will then summarize the resources that are available on campus and show how you can also use digital photography without owning specialized equipment.

T49. The central nervous system of the ascidian *Ciona intestinalis*

Sarah Stanley

Ascidians, also called tunicates, are a group of marine creatures that exhibit several of the defining chordate features of the vertebrate brain. They are thought to lie closest to the presumed ancestral group of all chordates, including vertebrates, from the body plan of their tiny tadpole larva, only several millimeters in length. Despite its size, the larva has a perfectly diminutive chordate central nervous system (CNS) with only several hundred cells in total. This makes the ascidian larva an attractive animal in which to perform detailed studies of a primitive and relatively simple CNS of the vertebrate type. 3-D computer reconstructions were made of all the CNS cells of individual larvae from the Southern Hemisphere, from serial sections. The cells within the CNS were classified by their nuclei and the appearance of their cytoplasm, as one of twelve possible types. Cell counts were compared with those from a previous study on larvae collected from the Northern Hemisphere. Several novel conclusions on larval morphology were drawn from this serial-section study and the resulting reconstructions. These data indicate a remarkable similarity in larval organization between individuals of this species.

P1. Continuous delayed match/non-match-to-sample: An odor-guided test of working memory in the rat

Boyer Winters and Richard E. Brown

Animal models of working memory have traditionally used a delayed matching-to-sample problem with visual or spatial stimuli. We have developed a more ethologically valid procedure to assess working memory in the rat using olfactory stimuli. This task utilizes a go-left/go-right procedure and thus eliminates both the ambiguity of null responses inherent in the go/no go olfactory learning paradigm and the problem of mediating behaviours which plague other putative tests of working memory such as the delayed matching-to-position task (Stanhope et al., 1995, *Psychopharmacology*, 122, 268-280). Five rats were trained on a continuous delayed match/non match-to-sample (CDM/NMS) task in an operant olfactometer with four different delays (3, 30, 60, and 120 sec) between the sample and choice odors. The percent correct in this task decreased with increasing delay, indicating a working memory function and suggesting that this procedure is a good model for testing drugs which influence short-term memory. When a stable baseline level of performance was reached, the effect of the central muscarinic antagonist scopolamine hydrobromide (HBr) was assessed. Many experiments suggest that scopolamine has a detrimental effect on learning and memory processes (Smith, 1988, *Brain Research Reviews*, 13, 103-118). However, other studies suggest that the effect of scopolamine may be due to something other than memory impairment (Herremans et al., 1996, *Brain Research*, 711, 102-111). We injected rats i.p. 30 minutes before testing on different days with one of five treatments: saline, 0.3 mg/kg scopolamine methylbromide (a peripherally acting muscarinic antagonist), 0.03 mg/kg scopolamine HBr, 0.1 mg/kg scopolamine HBr, and 0.3 mg/kg scopolamine HBr. There was a significant decrease in performance with increasing delays and a significant decrease in percent correct as the dose of scopolamine HBr was increased. There was no interaction between drug and delay. The general decrease in percent correct was interpreted as indicating an effect of scopolamine on a non-mnemonic process which influences performance in the CDM/NMS olfactory learning task. It is suggested that scopolamine induces a failure to attend to relevant information.

P2. Saccadic Reaction Times are Influenced Similarly by Previous Saccadic Metrics and Exogenous Cueing in Monkey

M.C. Dorris, T. Taylor, D.P. Munoz, R. Klein.

Human subjects are slower to initiate saccades to targets presented at previously exogenously-cued locations (inhibition of return - IOR). We extended this same finding to rhesus monkeys using an IOR paradigm. Then using IOR as a tool we studied the link between the mechanisms underlying saccade generation and exogenous attentional shifts. We repeated the same IOR experiment but instead of the first peripheral stimulus being an exogenous cue, it was the target for a saccade. The monkey was required to fixated a centrally presented fixation point which stepped out to a peripheral location, back to center, and back to a peripheral location. The location of the two peripheral targets were randomly varied so as to range from coincident to highly disparate. Instead of increased SRTs when the two peripheral stimuli were coincident, as was observed for the exogenously-cued IOR paradigm, SRTs were increased when the two peripheral stimuli were of equal amplitude but opposite directions with respect to the fixation point. Therefore, when two successive saccades had the same metrics (e.g., from a 10 leftward target to center, and center to a 10 rightward target: two successive 10 rightward saccades) SRTs were increased. These findings are interpreted in terms of shared processing of oculomotor programming and visuospatial attention on the same neural structure. To provide a direct confirmation this interpretation activity of single neurons in the superior colliculus of this monkey were recorded, and we hope to have some of these data analyzed in time for presentation of this poster.

P3. The Effects of Nicotine on Spatial and Non-spatial Expectancies in a Covert Orienting Task (or "Reduce Costs by Smoking")

Raymond M. Klein, Dalhousie U., Psychology
Fionnuala Murphy, Cambridge U., Psychiatry

Observers respond faster to targets presented at previously cued locations. Alzheimer's patients and monkeys with lesions of the basal forebrain cholinergic system show an increase in this cuing effect. We hypothesized that because nicotine is a cholinergic agonist it might have the opposite effect (reducing the cuing effect). 3 dose levels were achieved by testing casual smokers at each of three delays after smoking one cigarette. Peripheral flashes drew attention to the location where the size of a peripheral object was most likely to be changed, and participants performed a speeded discrimination (expansion/contraction). The performance decrement (cost) obtained when the target appeared at an uncued location was smallest in sessions run immediately after smoking (when nicotine levels would have been highest), suggesting that nicotine may increase the ease with which attention can be disengaged from a cued location. Our findings converge with other research supporting a role for the basal forebrain cholinergic system in visual orienting.

P4. Auditory and Visual Control of Fixation

Tracy L. Taylor & Raymond M. Klein

In the rostral pole of the superior colliculus (SC) are fixation cells that inhibit the remainder of the intermediate SC when a visual stimulus is being fixated. This inhibition is removed when the visual fixation stimulus is extinguished coincident with or prior to the onset of a peripheral target, resulting in facilitated saccadic reaction times (RTs). Because the SC also receives auditory input, the purpose of the present study was to determine whether this gap effect could also be obtained for saccade targets that follow the offset of a fixed auditory stimulus. Experiments 1 and 2 presented undergraduate subjects with three light emitting diodes (LEDs) mounted in front of speakers and spaced 10 degrees apart along the horizontal meridian. At the centre location, subjects were required to fixate an illuminated LED or else an auditory stimulus (noise in E1; a tone in E2). This fixated stimulus was offset simultaneous with or 200msec prior to the onset of a noise and/or LED presented equiprobably to the left or right. In E2, a condition was included for which the fixated stimulus remained present for the trial duration. The results demonstrated a facilitatory effect of removing a fixated auditory stimulus but only when the saccade target had a visual component (either visual or visual+auditory); the same was true for a fixated visual stimulus. When the saccade target was auditory, there was interference from the simultaneous offset of an auditory fixation, but no such interference from the offset of a visual fixation. This interference effect notwithstanding, the results demonstrate the similarity of visual and auditory stimuli for controlling fixation and suggest that examination of auditory input into the fixation cells of the rostral SC would be fruitful.

P5. Visual Orienting in Peri- and Extra-Personal Space by Patients with Stroke-Induced Parietal Damage

Bruno J. Losier and Raymond M. Klein

When patients with posterior right-hemisphere lesions perform Posner's visual orienting task a "disengage deficit" for uncued targets in contralesional space is consistently obtained. Converging evidence from single unit recording and human line bisection studies suggests that the parietal lobes preferentially process visual stimuli within grasping (peripersonal) space. This led us to hypothesize that the disengage deficit would be more severe in peripersonal than extrapersonal space, and in the lower visual field. Fifteen patients with posterior right-hemisphere lesion documented by CT scan (and normal controls) responded to validly, invalidly and neutrally cued target stimuli presented at the four corners of an imaginary square (8 deg across) at two distinct viewing distances (.280 m.=peripersonal and 2.24 m.=extrapersonal space), while maintaining central fixation. Cuing effects did not differ in peripersonal and extrapersonal space. A sub-sample of patients with parietal lesions showed greater deficits to both cued and uncued targets in the contralesional than ipsilesional visual field. For these patients, the disengage deficit was greater in the lower (contralesional) quadrant. Surprisingly, following contralesional cues performance was identical for targets at the cued and uncued locations within the contralesional field. A dissociation in the previous pattern of response in peripersonal and extrapersonal space was noted in some parietal patients. Our findings are discussed in light of Previc's proposal of a parietal specialization for the lower visual field, as well as Graves' suggestion that neglect is akin to the converse of blindsight (awareness without localization).

P6. Smell perception and pregnancy sickness

Farhad Dastur¹, Donald Brown², Kathy Harrigan³, Heather Wile³, & Richard Brown¹

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OBJECTIVE: To examine changes in smell perception (sensitivity, identification, qualitative ratings) during pregnancy and how those changes influence symptoms of "pregnancy sickness" (PS): food/odor aversions, nausea, retching, and vomiting.

DESIGN: A prospective, controlled, and longitudinal design.

SETTING: A quiet, ventilated testing room in the IWK-Grace Health Centre.

PARTICIPANTS: 20 healthy adult pregnant women and 20 healthy adult nonpregnant women not taking hormone-based contraceptives.

MAIN OUTCOME MEASURES: Olfactory detection threshold; olfactory identification score (UPSIT) Likert-scale ratings of odour preference, intensity, familiarity, nauseousness, palatability, and evocativeness; number of food aversions and cravings; energy intake (kJ); nausea (frequency, duration, and distress); retching (frequency and distress); and vomiting (frequency, quantity, and distress) over a 12-hour period.

RESULTS: A pilot study revealed that first trimester pregnant women had a lower olfactory threshold (i.e., greater olfactory sensitivity) than nonpregnant women ($t(8) = 2.46$, $p < .05$, two-tailed). More recent and comprehensive results will be presented.

CONCLUSIONS: These preliminary results are consistent with the clinical observation and the theoretical prediction that first trimester is associated by marked changes in olfactory perception, specifically increased olfactory sensitivity.

P7. The Temporally Extended Self in Young Children

Karen Skene & Chris Moore

Preschool children's awareness of self through time, or their Temporally Extended Self, was investigated in this experiment. Forty children aged 3;6-4;0 (years;months) completed a delayed self-recognition task, an episodic memory task, and a task to measure future-oriented prudence. Results indicated that most children were able to correctly label their video image as "me". However, children who showed delayed self-recognition scored higher on both memory and future-oriented prudence than children who did not demonstrate delayed self-recognition. We propose that awareness of the temporally extended self may develop first as an identity relation and then as a causal relation.

P8. Conditioned aversion to aqueous odours

Richard E. Brown, Timothy D. Moran and Burton M. Slotnick

Three experiments were conducted to further investigate the nuances of the conditioned odour aversion paradigm using aqueous odours. First, the effects of delay between the CS (0.05 % amyl acetate solution) and US (LiCl) in developing an aversion was investigated in a one bottle test with CS-US intervals of 0.25, 1.0, 2.0, 4.0, and 8.0 hr. The 0.25 and 1.0 hr delays developed the strongest aversions to amyl acetate. Experiment 2 used a two bottle test to examine whether the conditioned aversion would generalize to a novel aqueous odour and whether animals must consume the liquid to develop an aversion to the odour. When the odour presented during extinction (almond) differed from that presented during conditioning (amyl acetate) drinking was not suppressed, indicating that conditioning was specific to the odour used on the conditioning day. When access to the odorous liquid was prevented on the conditioning day by a wire-mesh barrier, an odour aversion was not learned. Experiment 3 investigated long term odour aversions. Twelve weeks after one conditioning trial to amyl acetate odour, drinking was still suppressed, indicating a long-term memory for the odour. These results support those of Slotnick et al (in press). They indicate that odour aversion learning is a powerful phenomenon, that it is odour specific, that there must be consumption of the liquid, and that there is a long term memory for the odour. These results indicate that some revision to Garcia's theory of conditioned aversions is required.

P9. To squeak or not to squeak: The relation between ultrasonic vocalization and behavioural immobility in the rat

Robert C. Barnet, Laura H. Corbit and Vincent M. LoLordo

Experiments with adult rats were conducted to explore predictions of a leading model of defense fear (Fanselow, 1994). In the present work, rats were exposed to footshock that occurred at a particular and consistent temporal location withing training sessions during which behavioural immobility ("freezing") and ultrasonic vocalizations (UVs) were recorded. Of interest, was the course of acquisition for the ability of contextual cues of the training chamber to elicit behavioural immobility and UVs. Fanselow's (1994) model assumes freezing to be controlled by the ventral periaqueductal gray matter of the midbrain (vPAG), and ultrasonic vocalization to be controlled by the dorsolateral periaqueductal gray (dlPAG). Activation of the dlPAG is thought to inhibit activation of vPAG. Accordingly, UVs and freezing should not co-occur and the model predicts increases in contextually-elicited freezing. Data from the present work challenge this prediction of the model. Benefits and limitations of using ultrasonic vocalization as a measure of learned fear are discussed.

P10. Construct validity of a yes/no recognition memory test used to predict and prevent global amnesia

Jeannette McGlone, S.E. Black and A. Sita

This investigation examined the construct validity of a yes/no recognition memory protocol routinely used to predict amnesia during an injection of intracarotid sodium amobarbital prior to a temporal lobectomy. One patient with global amnesia, two with severe but not global amnesia, and one with Wernicke's aphasia were administered the protocol without any drug injection. The globally amnesic patient scored below chance on the memory protocol, and 100% correct on the language protocol. Two cases of severe selective amnesia passed both the memory and language protocols. The aphasic passed the memory protocol, but made errors on the language protocol. Combined, these preliminary results indicate that this yes/no memory protocol has both convergent and divergent validity for differentiating global amnesia and aphasia.

P11. The Description of Personality from a Defense Mechanism Perspective: A Nomothetic and an Idiographic Approach

Michael MacGregor and Darcy Santor

Interest in defense mechanisms has increased, however, there is little consensus how defense mechanisms can be used to understand personality (nomothetically or idiographically). This paper discusses how an idiographic instrument can be used nomothetically with group norms to understand personality from a defense mechanism perspective.

The Defense-Q was used to assess 250 students (125 men) for the use of 25 defense mechanisms. The BFI-54 was used to assess the five factors of personality. When describing men and women, men had high use of Rationalization and low use of Acting Out. Women had high use of Rationalization and Passive-Aggression. When describing the five factors, individuals highest on Conscientiousness had high use of Pseudoaltruism and Reaction Formation. Analyses were conducted for each of the five factors.

Results are discussed in terms of group norms for idiographic measures of defense mechanisms and the relationship between defense mechanisms and personality.

P12. The Role of Objects in the Fixation Offset Effect

Ray Klein

Saccadic reaction time (SRT) is reduced when the fixation stimulus is removed immediately before the target appears, an effect that is thought to be mediated by intra-collicular inhibition of the motor programming machinery in the superior colliculus emanating from the fixation zone in the rostral pole. To explore the role of object properties in mediating this effect, observers fixated the intersection of a horizontal and a vertical line which bisected the display monitor while an outline 2.8° square, drawn in the same color and thickness as the lines, appeared 3.7° above the intersection. On 1/2 of the trials this square appeared solid and occluded the vertical line; on the remaining trials the outline square was merely added to the drawing. One of three types of events occurred simultaneously with (step), or 200 ms before (gap), a visual target (filled circles) that appeared randomly 5.7° to the left or right of fixation: 1) center disappear (the central 2.8° portion of the intersection disappeared) 2) square disappear (the square was removed from the display); 3) square to center (square moved from its position above the fixation to the center of the intersection). In the main experiment observers made saccadic eye movements to the targets. SRT on step and gap trials was facilitated equally (compared to square disappear) whether the intersection was removed (center disappear) or occluded (in the square to center condition). On step, but not gap, trials SRTs were delayed in the square to center condition. In a control experiment, where manual responses were made to indicate on which side the target appeared, the pattern of performance was quite different, indicating an oculomotor locus for the aforementioned effects which are interpreted in terms of neural systems involved in oculomotor programming.

P13. Object Influences on Illusory Line Motion

W. C. Schmidt and Raymond M. Klein

We sought to determine whether illusory line-motion would show object-level effects beyond those reported previously. Three experiments were conducted using ratings of ILM frequency and strength as dependent measures, and convergences was demonstrated with data collected by others using traditional measures of visual and cognitive processing.

